Subject: DATABASE MANAGEMENT SYSTEMS | **Type:** Previous Year Question Paper
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Mid-Semester Examination I - Question Paper (2025)

Part A: Short Answer Questions (5 x 2 = 10 Marks)

1. Define data abstraction and distinguish between physical, logical, and view levels.

2. State the difference between strong and weak entity sets with an example.

3. What is a candidate key? How does it differ from a primary key?

4. Explain the significance of the Cartesian product in relational algebra.

5. Write the SQL syntax for creating a unique constraint on multiple columns.

Part B: Descriptive Questions (3 x 10 = 30 Marks)

6. Design an ER diagram for a Hospital Management System showing all entities, attributes, primary keys, and relationship cardinalities.

7. Given relation R(A, B, C, D) with FDs {A -> B, B -> C, C -> D}, determine all candidate keys and verify if R is in 3NF or BCNF.

8. Explain basic Relational Algebra operators with suitable schema and queries.

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